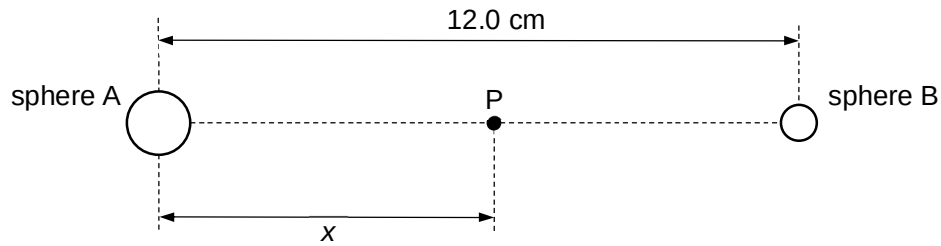


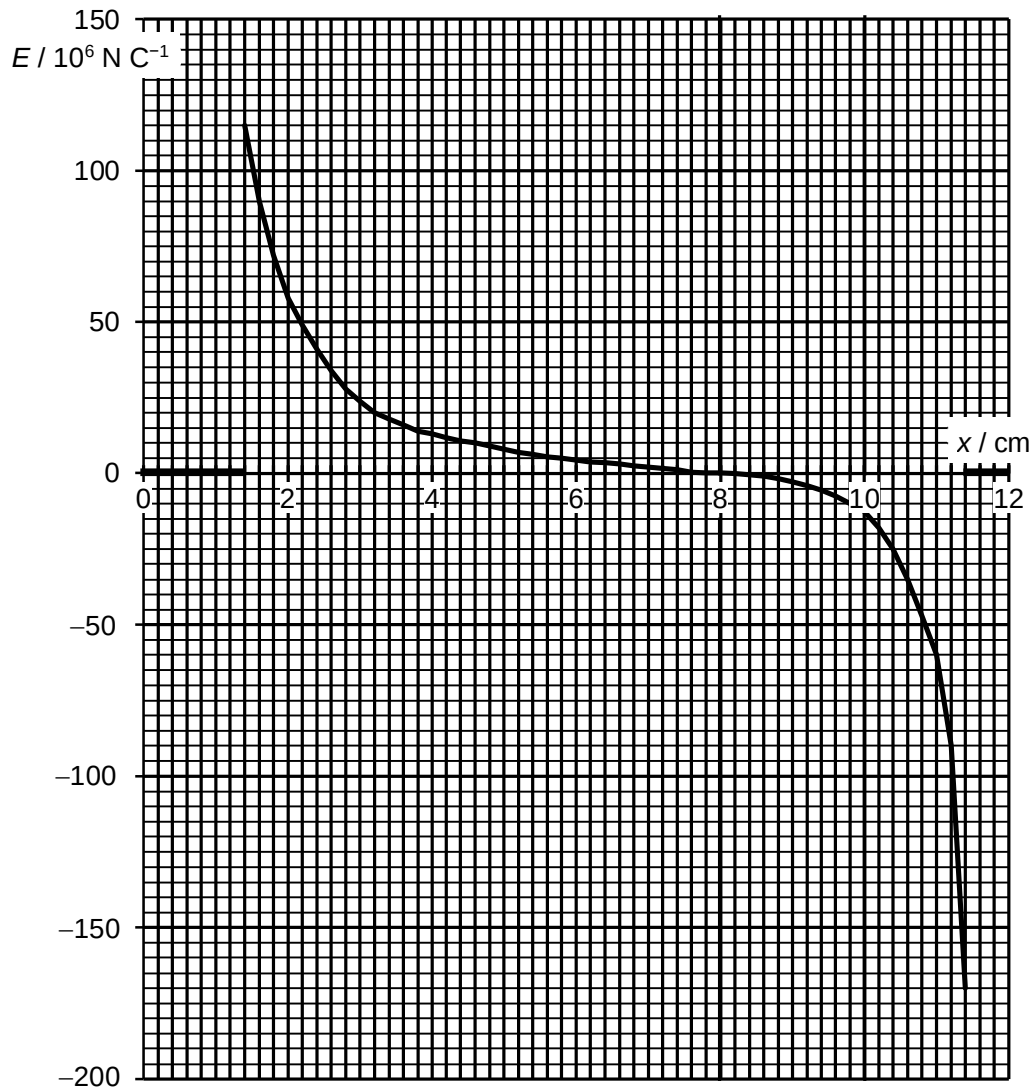
- 5 Two charged metal spheres A and B are situated in a vacuum. The distance between the centres of the spheres is 12.0 cm, as shown in Fig. 5.1.



**Fig. 5.1**

The charge on each sphere may be assumed to be a point charge at the centre of the sphere. Point P is a variable point that lies on the line joining the centres of the spheres and is distance  $x$  from the centre of sphere A.

The variation with distance  $x$  of the electric field strength  $E$  at point P is shown in Fig. 5.2.



**Fig. 5.2**

**(a)** State the evidence provided by Fig. 5.2 that the spheres are conductors.

.....  
 ..... [1]

**(b) (i)** If sphere A is positively charged, state and explain the polarity of sphere B.

.....  
 .....  
 .....  
 ..... [2]

**(ii)** Use Fig. 5.2 to determine the ratio  $\frac{\text{charge on sphere A}}{\text{charge on sphere B}}$ .

ratio = ..... [2]

**(c) (i)** State the relation between electric field strength  $E$  and electric potential  $V$ .

..... [1]

**(ii)** A point charge of  $-2.0 \mu\text{C}$  is moved by an external force from  $x = 2.0 \text{ cm}$  to  $x = 8.0 \text{ cm}$ , along the line joining the centres of the spheres.

Use Fig. 5.2 to estimate the work done by the external force.

work done = ..... J [2]

**6** Fig. 6.1 shows a resistor network connected to a 12 V battery of negligible internal